



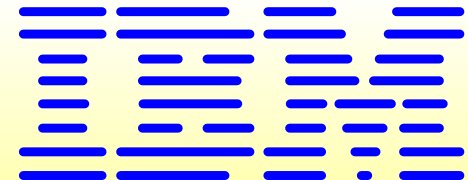
3018: DFSMS Basics: The ABC's of DFSMSrmm Part 1 of 2

Presented by:

Mike Wood
IBM Corporation
Warwick, UK

MikeW_Wood@UK.IBM.COM

A functional introduction to key RMM functions; how they work and what they deliver for your benefit. Learn how to define and manage the RMM resources and the options available to you.



NOTICES AND DISCLAIMERS

Copyright © 2005 by International Business Machines Corporation.

No part of this document may be reproduced or transmitted in any form without written permission from IBM Corporation.

Product information and data has been reviewed for accuracy as of the date of initial publication. Product information and data is subject to change without notice. This document could include technical inaccuracies or typographical errors. IBM may make improvements and/or changes in the product(s) and/or programs(s) described herein at any time without notice.

References in this document to IBM products, programs, or services does not imply that IBM intends to make such products, programs or services available in all countries in which IBM operates or does business. Consult your local IBM representative or IBM Business Partner for information about the product and services available in your area.

Any reference to an IBM Program Product in this document is not intended to state or imply that only that program product may be used. Any functionally equivalent program, that does not infringe IBM's intellectual property rights, may be used instead. It is the user's responsibility to evaluate and verify the operation of any non-IBM product, program or service.

THE INFORMATION PROVIDED IN THIS DOCUMENT IS DISTRIBUTED "AS IS" WITHOUT ANY WARRANTY, EITHER EXPRESS OR IMPLIED. IBM EXPRESSLY DISCLAIMS ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR INFRINGEMENT. IBM shall have no responsibility to update this information. IBM products are warranted according to the terms and conditions of the agreements (e.g., IBM Customer Agreement, Statement of Limited Warranty, International Program License Agreement, etc.) under which they are provided. IBM is not responsible for the performance or interoperability of any non-IBM products discussed herein.

The performance data contained herein was obtained in a controlled, isolated environment. Actual results that may be obtained in other operating environments may vary significantly. While IBM has reviewed each item for accuracy in a specific situation, there is no guarantee that the same or similar results will be obtained elsewhere.

The responsibility for use of this information or the implementation of any of these techniques is a customer responsibility and depends on the customer's or user's ability to evaluate and integrate them into their operating environment. Customers or users attempting to adapt these techniques to their own environments do so at their own risk. IN NO EVENT SHALL IBM BE LIABLE FOR ANY DAMAGE ARISING FROM THE USE OF THIS INFORMATION, INCLUDING BUT NOT LIMITED TO, LOSS OF DATA, BUSINESS INTERRUPTION, LOSS OF PROFIT OR LOSS OF OPPORTUNITY.

Information concerning non-IBM products was obtained from the suppliers of those products, their published announcements or other publicly available sources. IBM has not necessarily tested those products in connection with this publication and cannot confirm the accuracy of performance, compatibility or any other claims related to non-IBM products. Questions on the capabilities of non-IBM products should be addressed to the suppliers of those products. The provision of the information contained herein is not intended to, and does not, grant any right or license under any IBM patents or copyrights. Inquiries regarding patent or copyright licenses should be made, in writing, to:

IBM Director of Licensing
IBM Corporation
North Castle Drive
Armonk, NY 10504-1785

Trademarks



Removable Media Manager

The following terms are trademarks of International Business Machines Corporation in the United States, other countries, or both:

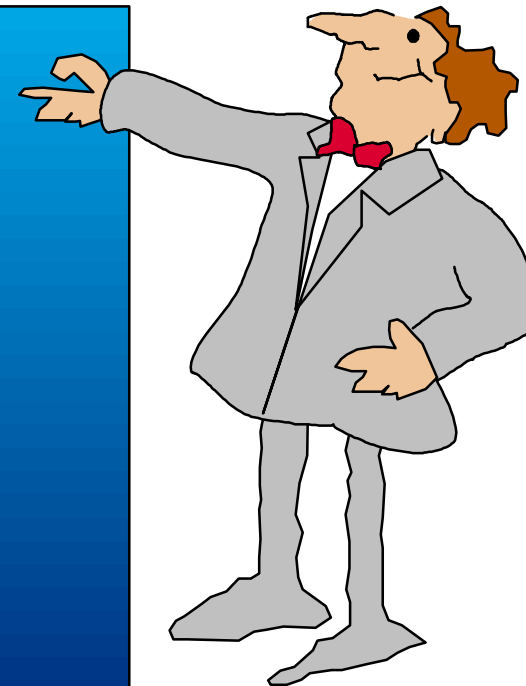
- DFSMSrmm
- IMS
- MVS
- OS/390
- RACF
- z/OS

Agenda



Removable Media Manager

- Introduction to RMM
- Resources managed by RMM
- Software Interfaces
- User/Program Interfaces
- Inventory Management
- Tools



our product ... DFSMSrmm



Removable Media Manager

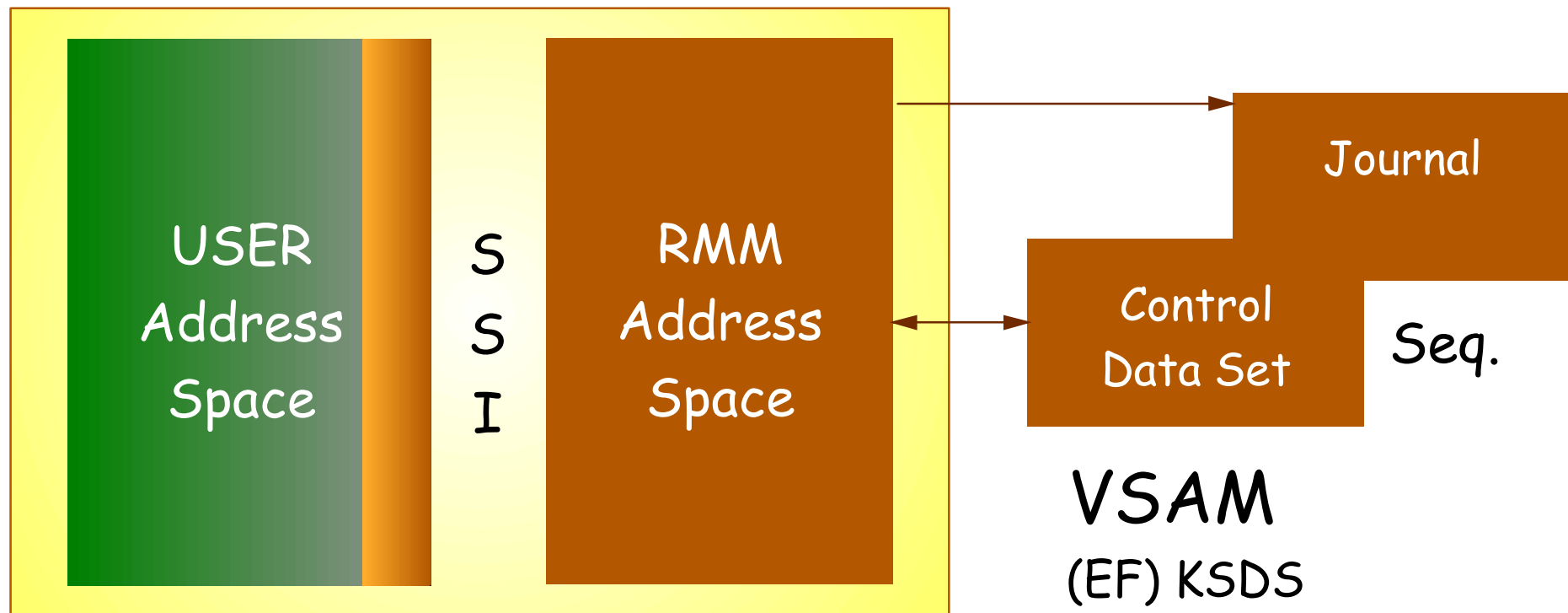
- Removable Media Manager
 - IBM's tape management system for z/OS
 - A complete, centralized and automated system
- Protecting tape data sets from accidental destruction
- Manage volume retention and movement
- Support and enhance system-managed tapes
- An Element of z/OS

Structure ...



Removable Media Manager

MVS



SSI = Subsystem Interface

VSAM
(EF) KSDS

(Extended Format)
Key Sequence Data Set

LSR

Local Shared Resource

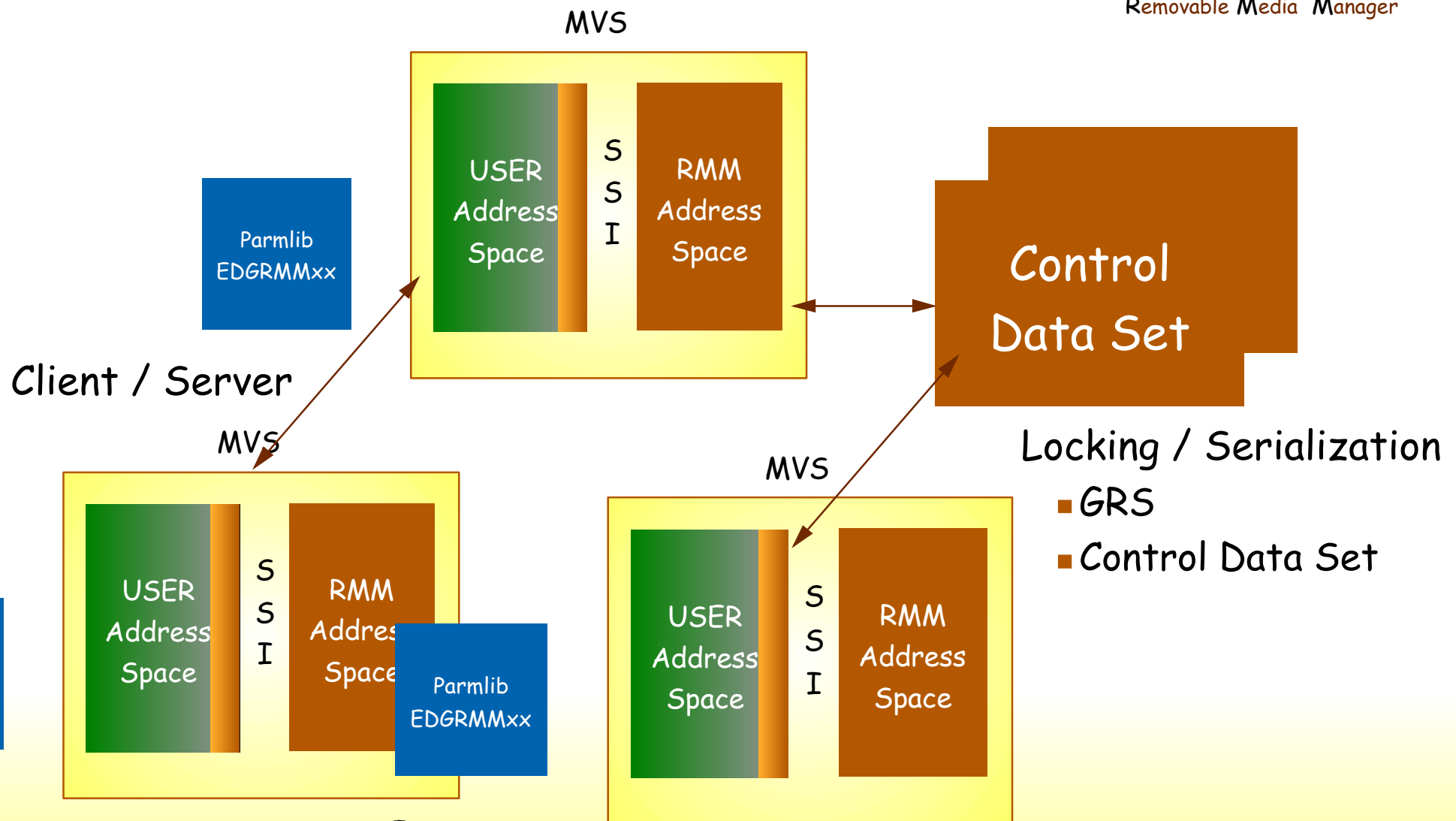


Copyright IBM Corporation 2006 (All Rights Reserved)

Multiple Systems ...



Removable Media Manager

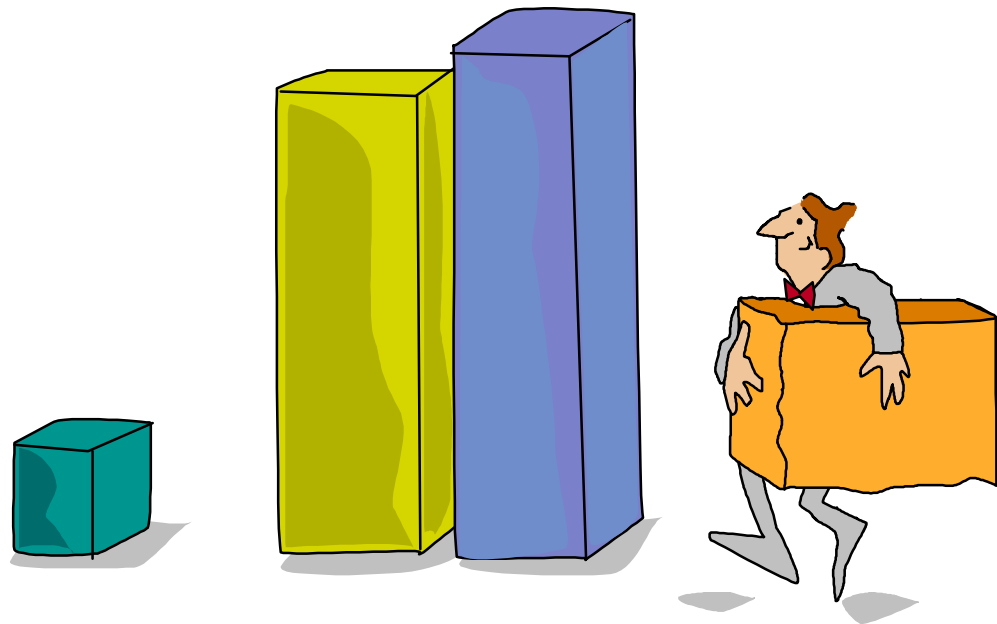


RMM managed resources ...



Removable Media Manager

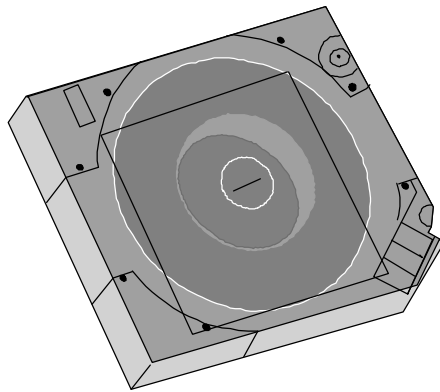
- Volume
- Data Set
- Owner
- Location
Rack / Bin
- Vital Record
Specification
- Software Product



Volume ... Types



Removable Media Manager



■ Physical Volume

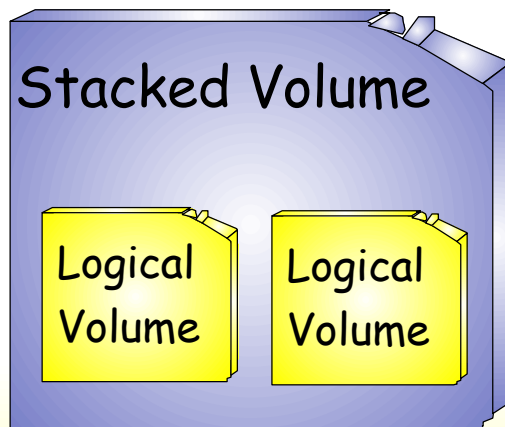
- WORM WWUCID

■ Logical Volume

- Used in a Virtual Tape Server
- Logical Volume / Virtual Volume

■ Stacked Volume

- Container for exported Logical Volumes (VTS)
- Box for Physical Volumes



Volume ...



Removable Media Manager

- Volume Status

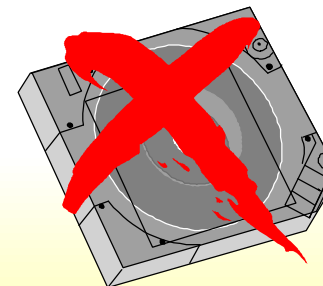
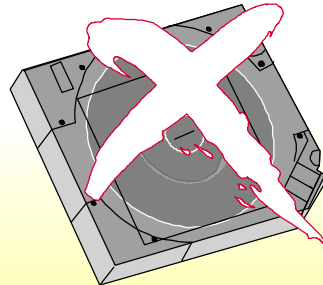
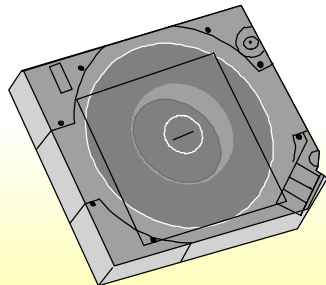
- Master
- Scratch
- User

- Expiration Date

- Availability

- Pending Release
- Vital Record retained

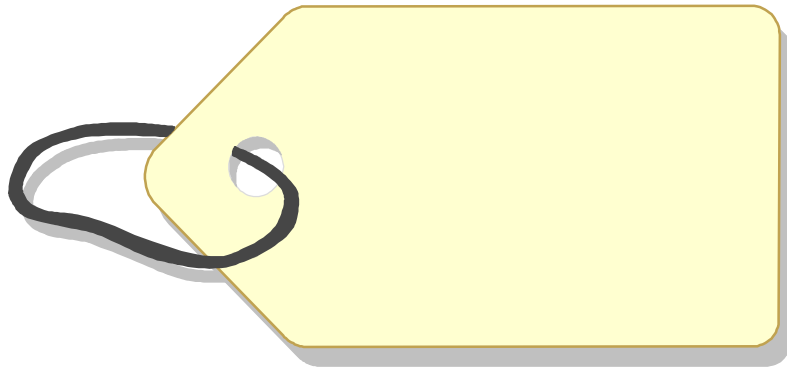
- Retention Date



Volume ...



Removable Media Manager



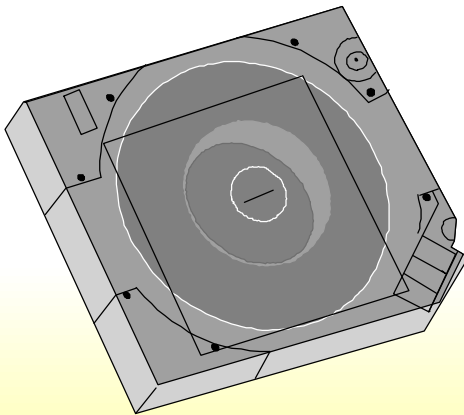
■ Label

- Standard Label
- ANSI Label
- No-Label

duplicate
VOL1
supported

■ Media Attributes

- Type
- Capacity
- Recording format
- Compaction

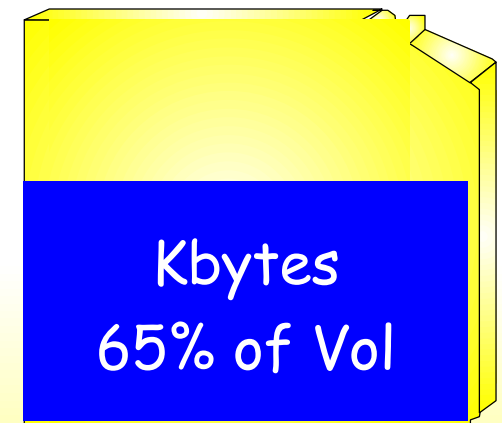
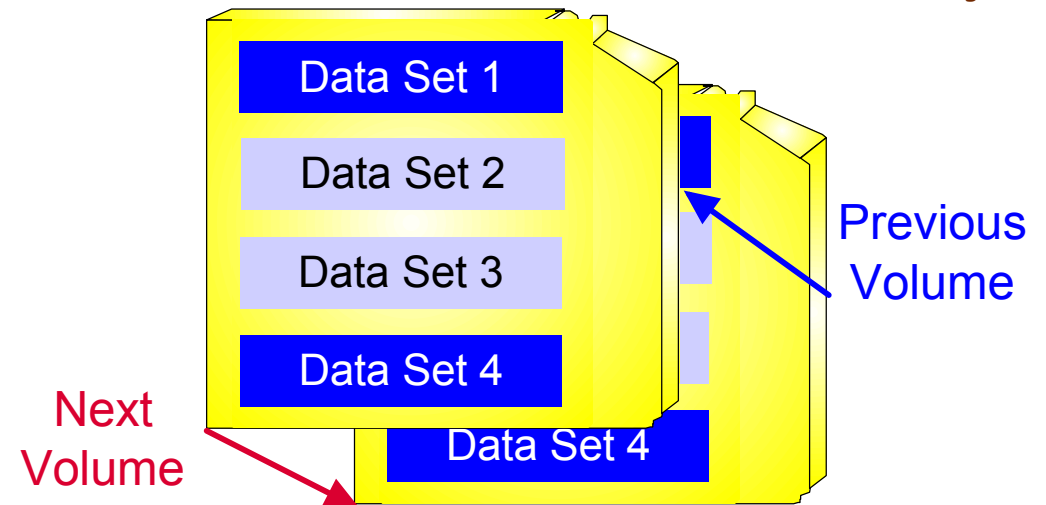


Volume ...



Removable Media Manager

- Data Sets on Volume
 - First DSN
 - Number of data sets
- Multi Volume Set
 - Next Volume
 - Previous Volume
 - Volume Sequence
- Volume usage
 - Application written
 - Hardware recorded



Volume ... Security



Removable Media Manager



- Security class
- Classification
- Owner
 - Owner Access
- Volume Access List

Volume ... Location



Removable Media Manager



- Current Location
- Home Location
- Required
- Destination Location
- Old Location
- Loan Location
- Rack / Bin
- In Transit Status

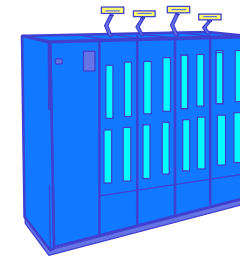
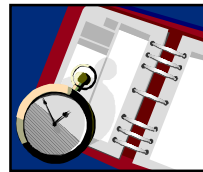
Volume ... Tracking



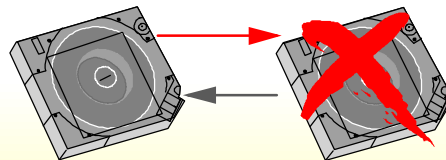
Removable Media Manager

- Last Change

- Date/Time
- Userid
- Systemid



- Status Change Date/Time



- Last Read Date
- Last Write Date

- Drive last used
- Last Move Date
- Write Mount

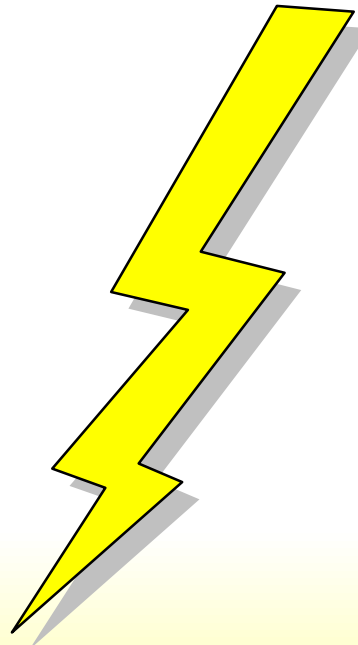
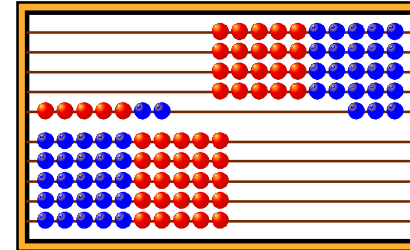
Count

Volume ... Tracking



Removable Media Manager

- Volume use count
- Volume Mount count
- Errors
 - Temporary
 - Read
 - Write
 - Permanent
 - Read
 - Write

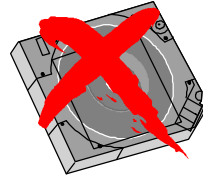


Volume ... Release Actions

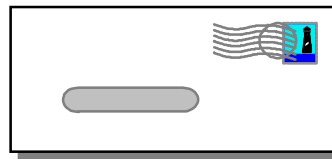


Removable Media Manager

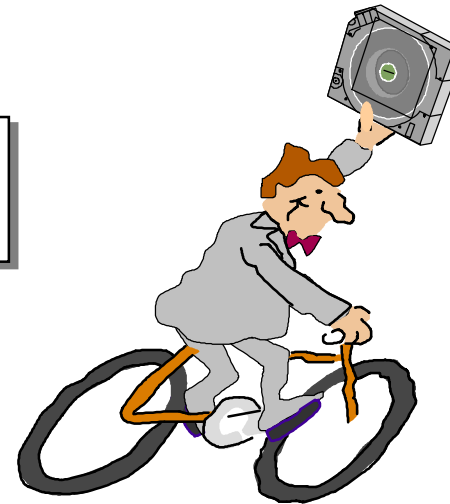
- Return to Scratch



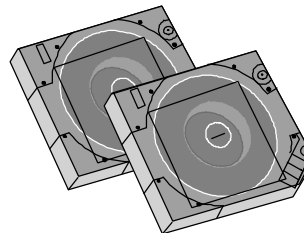
- Notify Owner



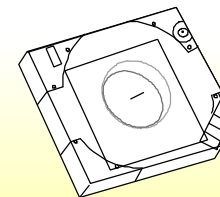
- Return to Owner



- Replace



- Init / Erase

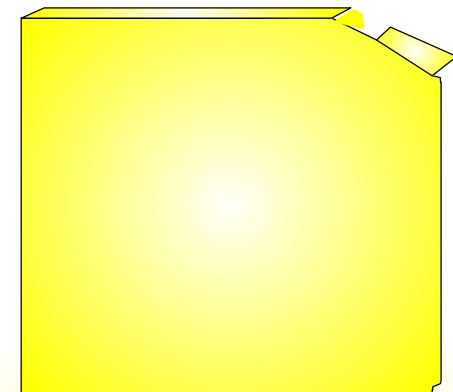
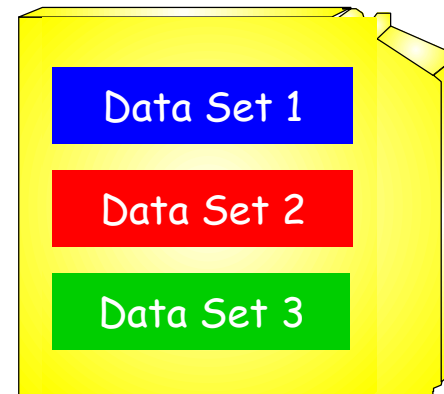


Data Set ...



Removable Media Manager

- Multiple Data Sets on one Volume

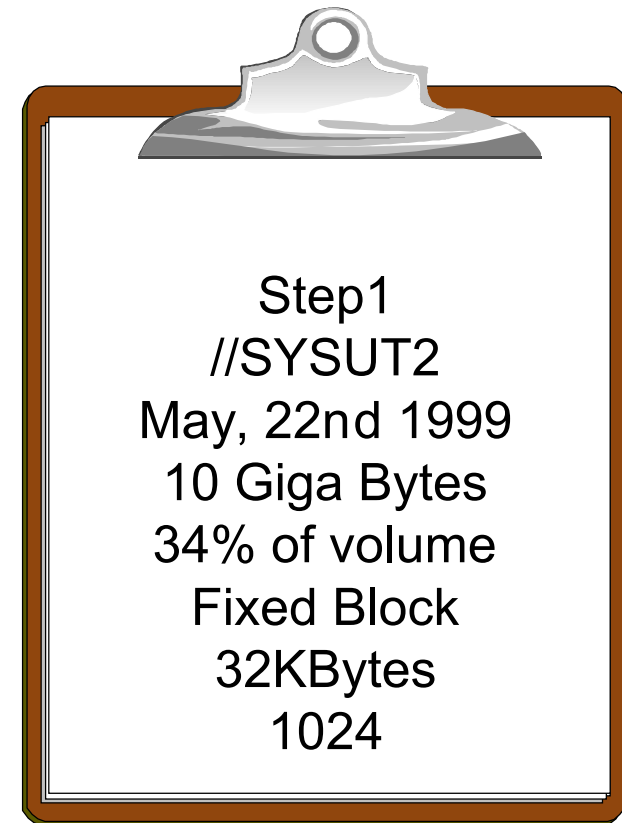


Data Set ... Attributes



Removable Media Manager

- Step Name / Program Name / DD NAME
- Create Date / Time
- Date last read / written
- Data Set Size
- Record Format
- Block Size
- Logical Record Length
- EXPDT and Original EXPDT

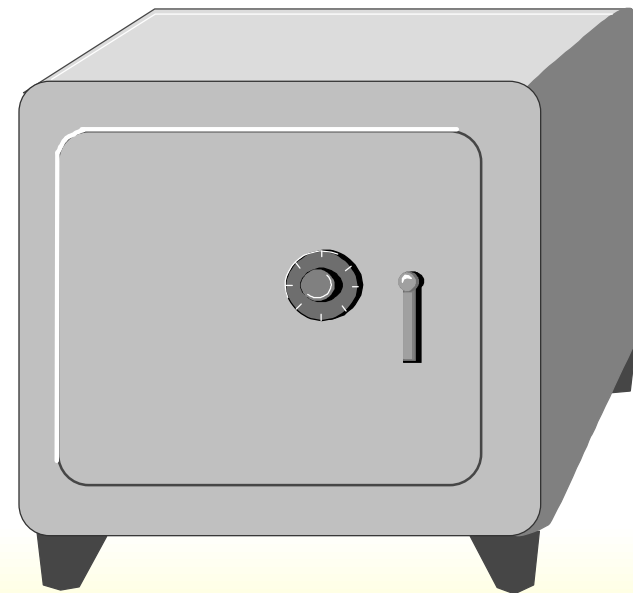


Data Set ... Vital Record



Removable Media Manager

- Vital Record Status
- Matching Vital Record Specification
- Current VRS Sub-Chain
 - name
 - date
- Retention Date

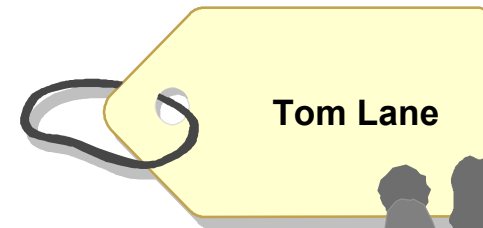


Owner ...



Removable Media Manager

- Name, Department, etc.
- Address
- E-mail

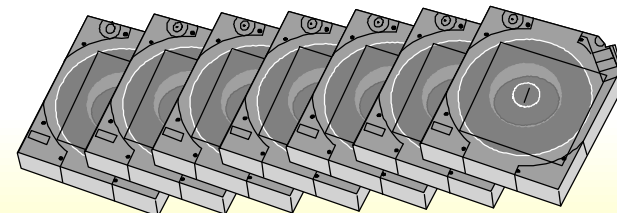


Dept. M44A



Rita Road, Tucson Az.

- List of owned Volumes

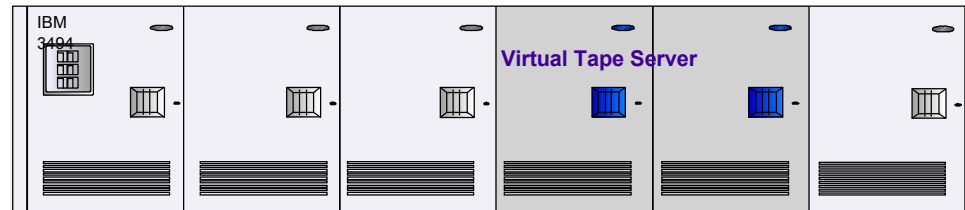


Location ...

System-Managed Tape Libraries



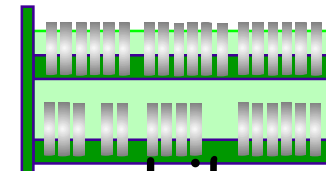
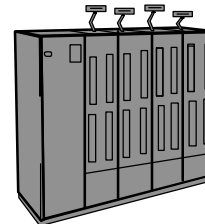
Removable Media Manager



- Automated

- IBM 3494 Tape Library Dataserver
- IBM Virtual Tape Server

- Manual



- A set of tape drives defined as a Library

- Location Name: `library_name`

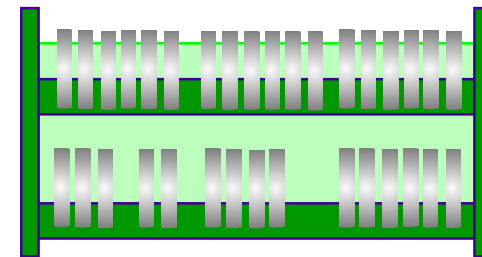
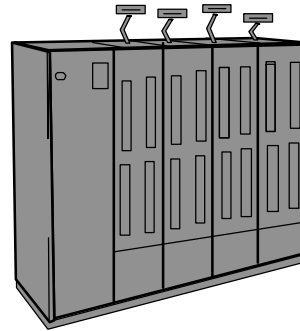
Location...

Non System-Managed Tape Lib.



Removable Media Manager

- Traditional Tape Library



- Other Robots
- Location Name: SHELF
: Any Store Name
- Rack management supported

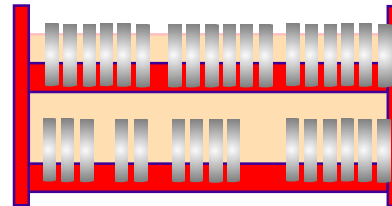
Location... Storage Location



Removable Media Manager

■ Management

- No Bins
- Bins
- Media name

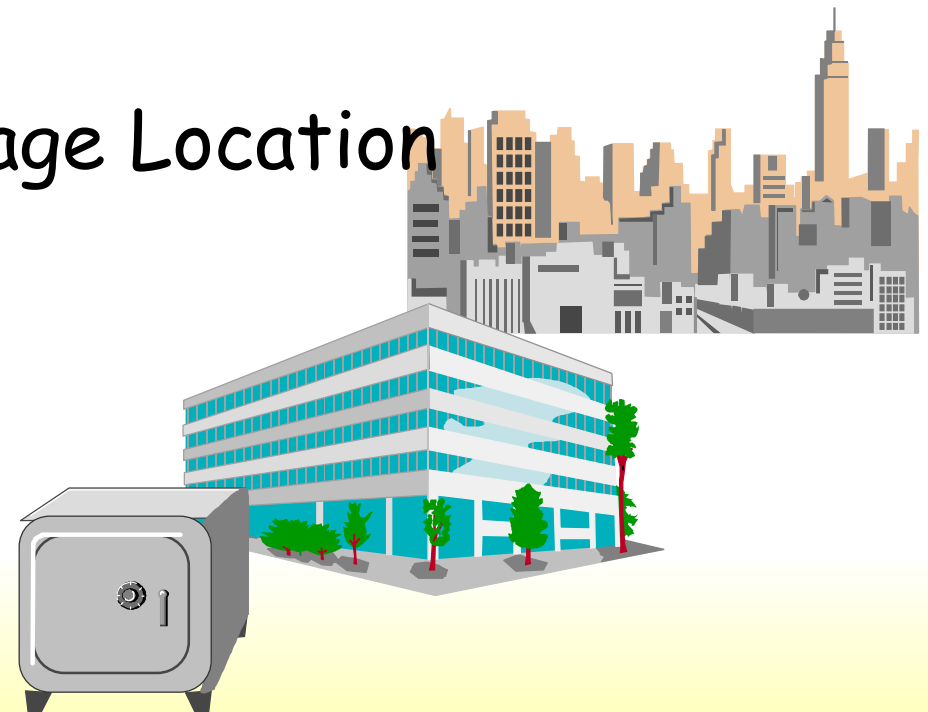


■ Installation defined Storage Location

- Unlimited number, any name

■ Built-in Storage Locations

- Location Name: LOCAL
- Location Name: REMOTE
- Location Name: DISTANT



Location... Shelf Space

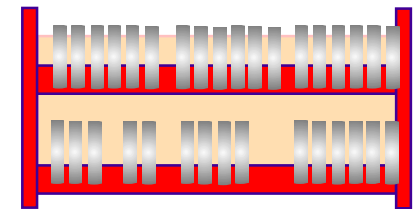


Removable Media Manager

■ Rack Number

- Optional for any Volume
- Assigned slot for volume storage
- Can be freed and re-assigned via commands
- Grouped into pools using VLPOOL PREFIX(pref*)

■ Selected using MEDIANAME



■ Bin Numbers

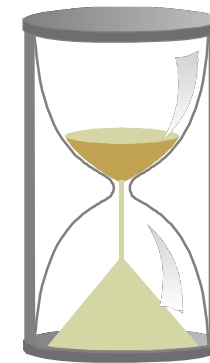
- Optional by Storage Location
- Assigned at start of a move
- Reusable either: on CMOVE out of Storage Location
- or: at Start of Move out of Storage Location

Vital Record Specification



Removable Media Manager

- VRS's are Policies to ...

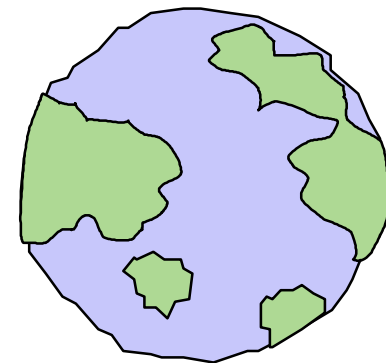


- VRSEL

- NEW/OLD



- determine Volume's required Location

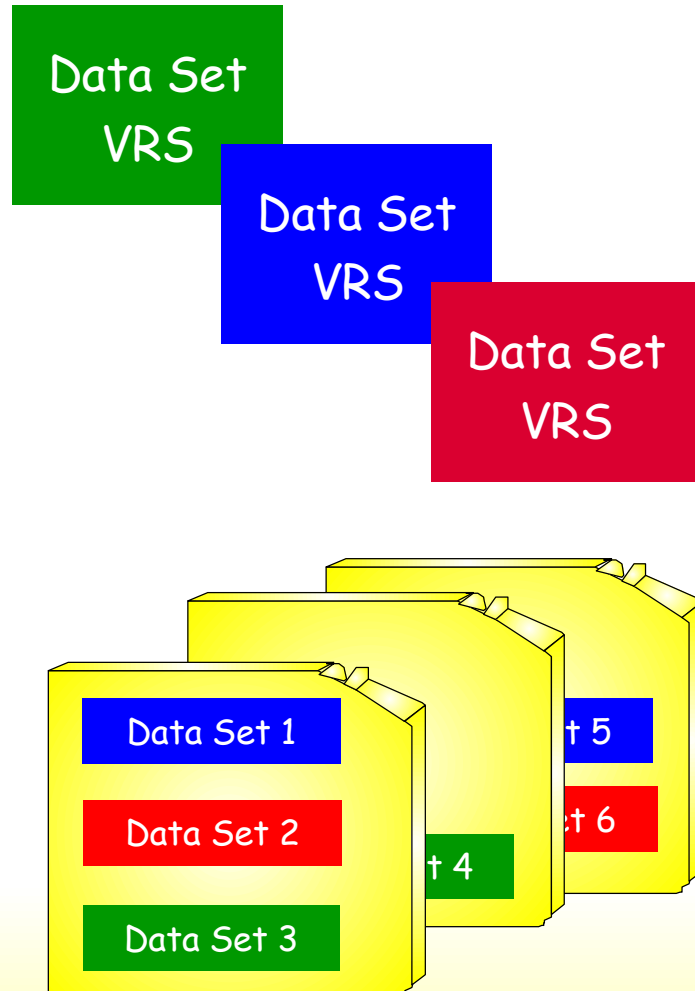


- Set Release Options

Vital Record Specification



Removable Media Manager



■ Data Set VRS

■ Select Data Sets by:

- OPEN flag
- ABEND flag
- Management Class
- Management Value
- Data Set Name
- Jobname
- Generation Data Group
- Generic Masks supported
 - Plus - Pseudo GDG

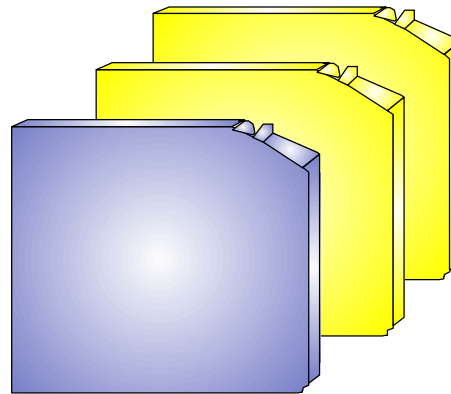
Vital Record Specification



Removable Media Manager

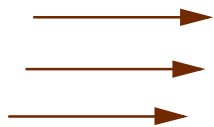
Volume
VRS

Volume
VRS



■ Volume VRS

- Specific
- Generic



Name
VRS

Name
VRS

Name
VRS

■ Name VRS's

- Location only
- Retention & Location

Vital Record Specification

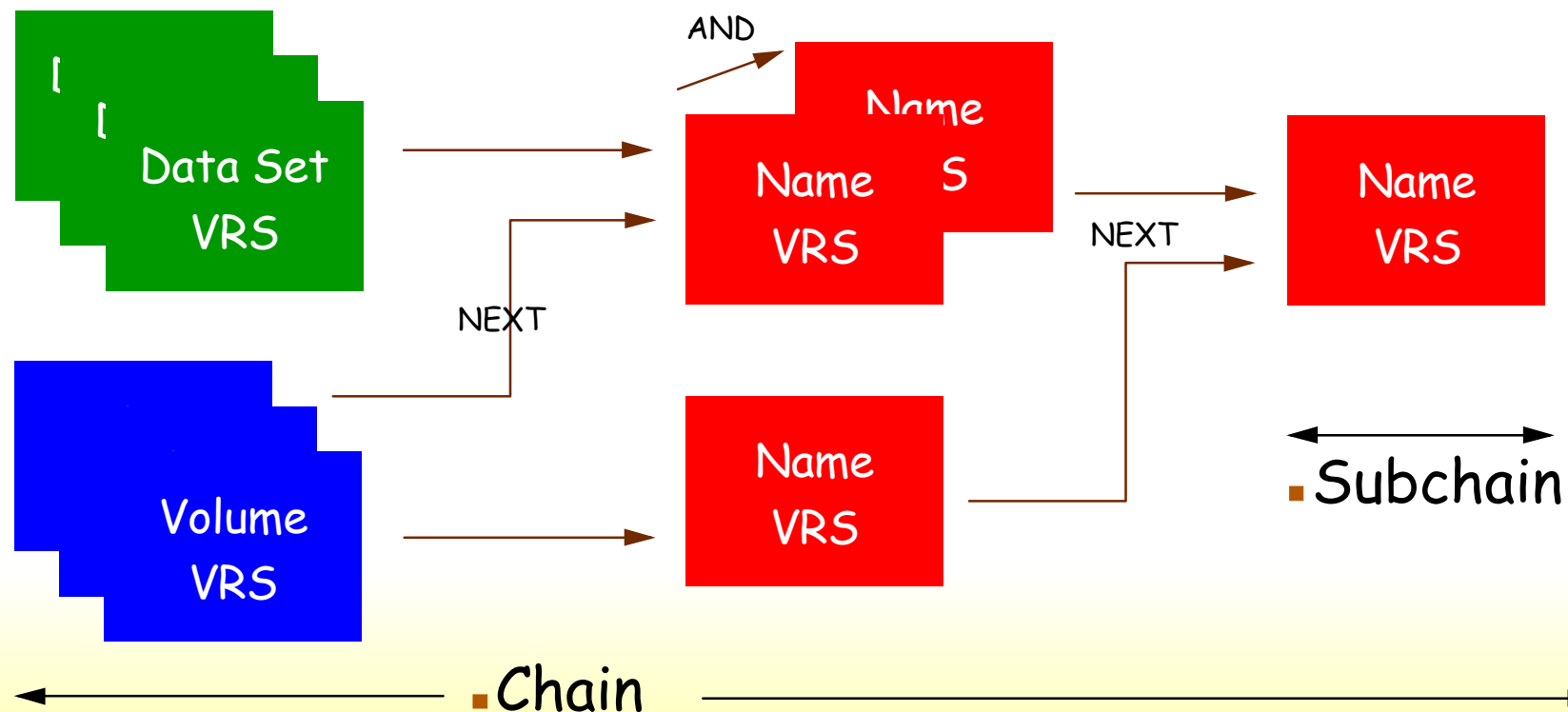


Removable Media Manager

■ Data Set VRS

■ Volume VRS

■ Name VRS



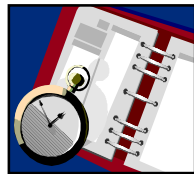


Removable Media Manager

Vital Record Specification

Retention Types

- Specific Date



- Days since creation
- Days since last reference
- Extra days



- Cycles
- By Days Cycle



- Forever

- While Cataloged
- Until Expired

Catalog

JCL



Vital Record Specification

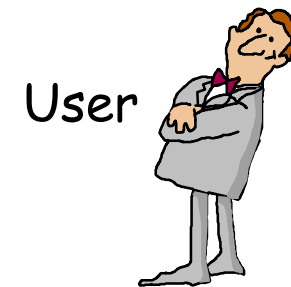


Removable Media Manager



Administrator

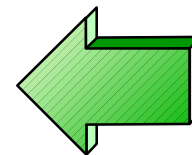
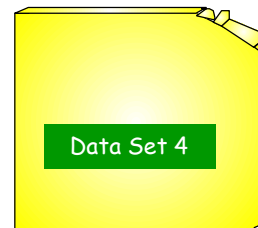
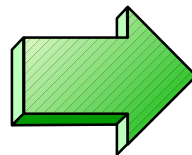
- Two Policies manage one data set



User



Primary VRS



Secondary VRS



Location



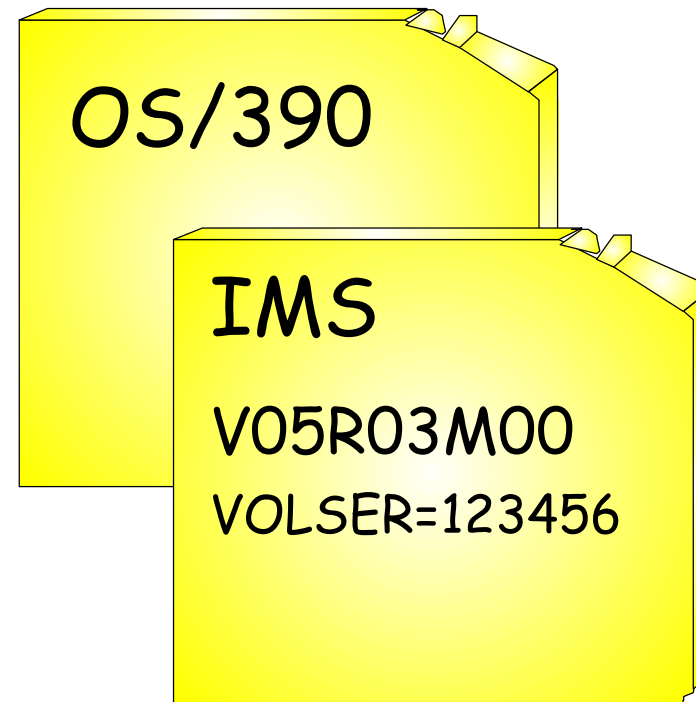
Retention

RMM managed resources ...



Removable Media Manager

- Volume
- Data Set
- Owner
- Location
 - Rack / Bin
- Vital Record
 - Specification
- **Software Product**

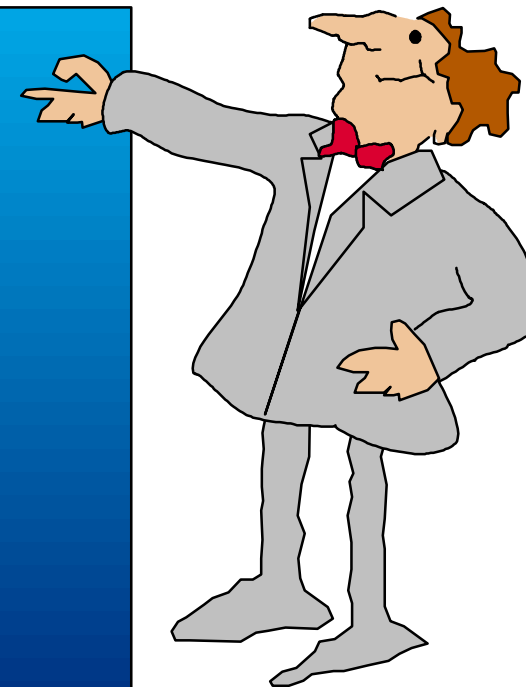


Agenda for Part II



Removable Media Manager

- Introduction to RMM
- Resources managed by RMM
- Software Interfaces
- User/Program Interfaces
- Inventory Management
- Tools



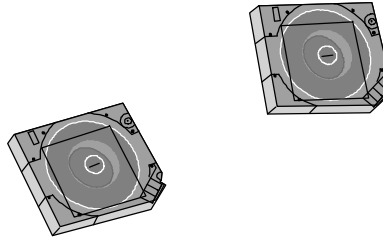
Summary



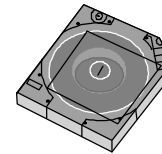
Removable Media Manager

Protect
Data on Tape

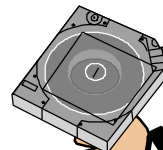
Enforce
Security



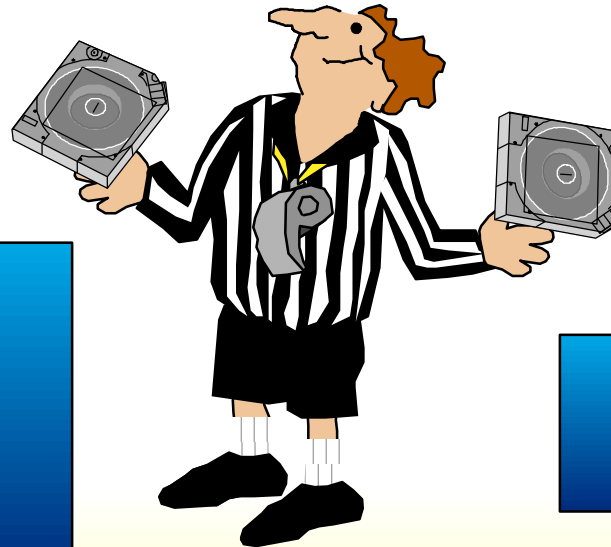
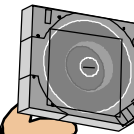
Control
Retention



Support
Automation



Manage
Vaulting



Thanks for your interest
in
DFSMSrmm ...

See you for Part II

Removable Media Manager



Copyright IBM Corporation 2006 (All Rights Reserved)